

Collecting Data from the Web

The Methodological Foundations of AI Screening

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instats Seminar

Thursday, June 11, 2026

The Screening Decision as a Classification Problem

Abstract screening is formally a **binary classification task** (relevant vs. not relevant) with highly asymmetric error costs.

False Negative

Missing a relevant study.

Cost: Categorically high. It biases the final synthesis and compromises the systematic review's integrity.

False Positive

Including an irrelevant study.

Cost: Low to moderate. The study will simply be excluded later during the full-text screening phase.

Implication: The objective function for screening must prioritise **recall** (sensitivity) over **precision**.

Sensitivity, Specificity, and the Recall–Precision Trade-off

Because most retrieved corpora are highly imbalanced (e.g., 5% relevant, 95% irrelevant), overall accuracy is a misleading metric.

Metric	Definition in Screening Context
Recall (Sensitivity)	Proportion of truly relevant papers correctly identified.
Precision	Proportion of identified papers that are actually relevant.
Specificity	Proportion of truly irrelevant papers correctly excluded.

The Operating Point

Systematic reviewers must deliberately set a **high-recall, low-precision operating point** on the ROC curve, accepting a large number of false positives to guarantee that no relevant studies are missed.

Inclusion Criteria as a Formal Decision Rule

A rigorous screening protocol translates PICO/SPIDER into a conjunction of necessary conditions. All conditions must be satisfied for inclusion.

A critical methodological distinction:

- **Eligibility criteria:** Does the paper address the correct population, intervention, and outcome? (Assessed at Day 2 screening).
- **Quality criteria:** Is the sample size sufficient? Is the risk of bias low? (Assessed at Day 3 extraction).

Methodological Error

Conflating eligibility and quality at the abstract screening stage is a common error. Quality appraisal requires reading the full text. Screening should assess *relevance*, not *validity*.

The Statistical Basis of Text Vectorisation (TF-IDF)

Active learning requires converting text into a numerical matrix. The standard approach is **Term Frequency–Inverse Document Frequency (TF-IDF)**.

The weight $W_{i,j}$ of term i in document j is defined as:

$$W_{i,j} = \text{TF}_{i,j} \times \log \left(\frac{N}{\text{DF}_i} \right)$$

where N is the total number of documents and DF_i is the number of documents containing term i .

Why it works:

- Rare, domain-specific terms (e.g., *echocardiography*) receive high weights.
- Common words (e.g., *study*, *analysis*) are down-weighted.
- This allows the classifier to locate papers in a high-dimensional vector space (Salton & McGill, 1983).

Active Learning: Query Strategies

Active learning is not a single algorithm; it is a paradigm where the model queries the user to label specific data points. The choice of **query strategy** dictates efficiency.

- **Uncertainty Sampling:** Query the instance the model is least certain about (e.g., predicted probability ≈ 0.5).
- **Query-by-Committee:** Query the instance on which an ensemble of models disagrees most.
- **Expected Model Change:** Query the instance that would most alter the model's weights if labelled.

Note: In systematic reviews, a variant of uncertainty sampling is typically used to rapidly surface the most likely relevant papers (Van de Schoot et al., 2021).

Stopping Rules and the WSS Metric

When using active learning, how do you know when to stop screening?

The standard performance metric is **Work Saved over Sampling (WSS@95)** (Cormack & Grossman, 2016):

- The proportion of the corpus that does *not* need to be screened to achieve 95% recall, compared to random screening.

The Stopping Dilemma

There is no universally agreed stopping rule. The tension is between efficiency (stopping early) and completeness (screening everything). In practice, reviewers often stop when a consecutive sequence of N papers (e.g., 50) are all classified as irrelevant.

LLMs as Classifiers: Theoretical Grounding

There are three paradigms for applying Large Language Models to classification:

Paradigm	Mechanism
Zero-shot	The model applies criteria from the prompt alone, with no labelled examples.
Few-shot	The prompt includes 3–5 labelled examples (in-context learning) to calibrate the model.
Fine-tuned	The model's internal weights are updated on a domain-specific labelled dataset.

Trade-offs: Zero-shot requires no training data but relies entirely on prompt clarity. Fine-tuning offers the highest performance on specialised text but requires significant computational resources and pre-labelled data.

The Reliability Problem: Hallucination and Confabulation

When using LLMs for screening, we must distinguish between two failure modes.

Hallucination

Generating plausible but factually incorrect content. In screening, this means citing criteria not present in the prompt, or misattributing data to the abstract.

Confabulation

Generating internally consistent but externally false reasoning. The model applies the criteria logically, but misunderstands the methodological context of the paper.

Mitigation

Demanding a structured justification in the prompt forces the model to articulate its reasoning, allowing the human reviewer to spot-check for these failure modes.

Prompt Sensitivity and the Reproducibility Problem

Empirical evidence shows that LLM outputs are sensitive to superficial prompt variations—word order, punctuation, and phrasing—that should be semantically equivalent (Mizrahi et al., 2024).

Implications for Systematic Reviews:

- Two researchers using slightly different prompt formulations may obtain systematically different screening decisions from the same model.
- LLM outputs are not deterministic; temperature settings must be fixed (e.g., $T = 0$).

The Solution: Pre-register prompts and report them verbatim in the methods section. Treat the prompt as a piece of scientific apparatus.

Measuring Agreement: Cohen's Kappa

To validate AI screening, we must measure inter-rater reliability. The standard metric is **Cohen's Kappa** (κ):

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

where P_o is observed agreement and P_e is expected agreement under the assumption of independence.

Interpretation (Landis & Koch, 1977):

- < 0.00 : Poor
- $0.00 - 0.20$: Slight
- $0.21 - 0.40$: Fair
- $0.41 - 0.60$: Moderate
- $0.61 - 0.80$: Substantial
- $0.81 - 1.00$: Almost perfect

For systematic reviews, $\kappa < 0.60$ generally requires criteria revision.

The Prevalence Paradox and Kappa's Limitations

Kappa has a well-documented mathematical vulnerability: it is highly sensitive to the prevalence of the positive class (Finstein & Cicchetti, 1990).

The Paradox

When the proportion of included studies is very low (e.g., 2%), Kappa can be misleadingly low even when absolute percentage agreement is 98%. The high expected agreement (P_e) artificially depresses κ .

Best Practice: Report Kappa alongside the **positive agreement (PA+)** and **negative agreement (PA-)** statistics. This provides a transparent view of where the raters diverge.

Validating AI Decisions Against Human Gold Standards

Validation is not adjudication. Validation measures agreement on a sample; adjudication resolves disagreements on the full corpus.

The Validation Protocol:

- 1 Select a stratified random sample of the corpus (e.g., 10% of AI-included, 10% of AI-excluded).
- 2 A human reviewer independently applies the criteria to this sample, blinded to the AI's decisions.
- 3 Compute κ , PA+, and PA-.
- 4 If $\kappa > 0.60$, proceed. If not, refine the prompt/criteria and re-run.

Note: The sample size must be large enough to yield a statistically meaningful confidence interval for Kappa.

Bias, Coverage, and External Validity

API-based retrieval inherits the structural biases of the published literature.

The File-Drawer Problem (Rosenthal, 1979):

- Studies with null or negative results are less likely to be published.
- APIs index published literature, meaning the retrieved corpus is already systematically biased toward positive findings.

Mitigation: API retrieval must often be supplemented with **grey literature** searches (theses, government reports, clinical trial registries) to capture unpublished null results.

Language, Geographic, and Indexing Bias

Beyond publication bias, three structural biases affect API-retrieved corpora:

- **Language bias:** Studies published in English are heavily over-represented in OpenAlex and Crossref relative to their share of global research output.
- **Geographic bias:** Research from high-income countries is over-represented due to institutional funding and publishing infrastructure.
- **Indexing bias:** Journals indexed in major databases are not a random sample of the literature; they skew toward established, high-impact venues.

These biases constrain the generalisability of the review's conclusions and must be explicitly acknowledged in the limitations section.

Reporting Standards: The PRISMA 2020 Flow Diagram

The PRISMA 2020 statement (Page et al., 2021) updated the reporting requirements for the flow of information through the review phases.

Key requirements for API-based reviews:

- Report the number of records identified through database/API searching separately from those identified through "other methods" (e.g., citation chasing).
- Report the number of duplicates removed prior to screening.
- Report the number of records excluded at the abstract stage.
- **Crucially:** Report the specific reasons for exclusion at the full-text stage.

Reporting AI Use: The Emerging Consensus

As AI tools enter the systematic review workflow, reporting standards are adapting (e.g., the PRISMA-AI extension).

Mandatory Disclosures (Flemyng et al., 2025)

- 1 The tool name, model architecture, and version (e.g., Llama 3 8B).
- 2 The exact prompt text used for zero-shot classification.
- 3 The validation procedure and inter-rater reliability metrics.
- 4 The human oversight protocol (e.g., human-in-the-loop vs fully automated).

The distinction between "AI-assisted" (human verified) and "AI-conducted" (fully automated) screening is critical for the credibility of the review.

The Included Studies Dataset as a Research Object

The output of Day 2 is not merely a list of papers; it is a **methodological record** with its own provenance and structure.

Data Lineage: Every record in the included set must be traceable back to a specific API query, a specific deduplication decision, and a specific screening decision.

Column	Function
Relevance_Score	Output of the Active Learning model.
LLM_Decision	Zero-shot binary classification (Include/Exclude).
LLM_Justification	The model's articulated reasoning.
Human_Verification	Record of human spot-checking or adjudication.

From Screening Justification to Data Extraction

The LLM justification field in the transparency log serves as a bridge to Day 3 data extraction.

Screening-Informed Extraction

A well-formed justification identifies which PICO elements were present in the abstract. This is essentially **proto-extraction**. By using the justification text to pre-populate the data extraction template, reviewers reduce cognitive load and improve consistency during the Day 3 synthesis phase.

The quality of the screening prompt directly determines the quality of the extraction data.

The Human-in-the-Loop Principle: A Methodological Commitment

We conclude Day 2 by reaffirming the core methodological commitment of AI-assisted systematic reviews.

Three levels of human involvement:

- 1 **Human as final arbiter:** Reviews all AI decisions (highest rigor, lowest efficiency).
- 2 **Human as spot-checker:** Reviews a statistically valid sample (the emerging standard).
- 3 **Human as exception handler:** Reviews only cases flagged by the AI as uncertain.

The goal is to leverage AI for scale and speed, while retaining human judgment for validity and accountability.